FORMAL MODIFICATIONS ARE STABLE UNDER BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the counted statement is the author lemma environment labelled lemma-base-change-formal-modification; the proof environment immediately following it is its own proof

Editorial note. The label lemma-base-change-formal-modification names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- the definition of a formal modification via properness, rig-etale-ness, and rig-surjectivity including diagonals
- the three base change lemmas of the same chapter: base-change-proper (Formal Spaces), base-change-rig-etale, and base-change-rig-surjective
- the functoriality of the /T completion construction under base change, so that the diagonal of the base change is the base change of the diagonal by an adic morphism

Source reference. Original lemma 0GBT: sources/stacks-chow/tags.txt, source lines 20383–20383. Prerequisite pointer invoked at proof line(s) 6376; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GD1: sources/stacks-restricted/pinned-original-source.tex, source lines 5125–5125. Prerequisite pointer invoked at proof line(s) 6378; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0AQS: sources/stacks-restricted/pinned-original-source.tex, source lines 5324–5324. Prerequisite pointer invoked at proof line(s) 6380; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of locally Noetherian formal algebraic spaces over S which is a formal modification. Then for any adic morphism $Y' \to Y$ of locally Noetherian formal algebraic spaces, the base change $f' : X \times_Y Y' \to Y'$ is a formal modification.*

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Proof. The morphism f' is proper by Formal Spaces, Lemma 0GBT. The morphism f' is rig-etale by Lemma 0GD1. Then morphism f' is rig-surjective by Lemma 0AQS. Set $X' = X \times'_Y$. The morphism $\Delta_{f'}$ is the base change of Δ_f by the adic morphism $X' \times_{Y'} X' \to X \times_Y X$. Hence $\Delta_{f'}$ is rig-surjective by Lemma 0AQS. $\square$